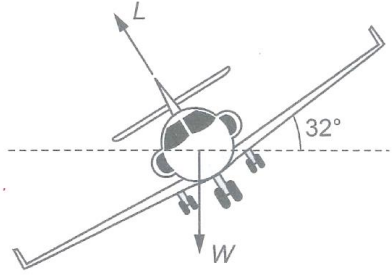
7 A small aircraft flies at speed v in a horizontal circle of radius r. To do so, the plane of its wings must be at an angle of 32° to the horizontal as shown. The directions of the two forces in the vertical plane acting on the aircraft, weight W and lift L, are also shown.



At which angle to the horizontal must the aircraft fly in a horizontal circle of radius $\frac{r}{2}$ at a speed of $2v$?

A 17° **B** 51° **C** 68° **D** 79°